

Markscheme

Mathematics: analysis and approaches

Practice question bank

18. (a)

Consider the function $f(x) = \frac{4x + 32}{(x + 6)(x - 2)}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x + 6} + \frac{B}{x - 2}$.

$$4x + 32 = A(x - 2) + B(x + 6) \quad (M1)$$

$$\text{substituting } x = -6: 8 = -8A \Rightarrow A = -1 \quad \mathbf{A1}$$

$$\text{substituting } x = 2: 40 = 8B \Rightarrow B = 5 \quad \mathbf{A1}$$

[3 marks]